

Name

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Activity - Recursion

Algorithm - AI 4-1

$$\textcircled{1} \quad T(n) = 5T(n/5) + n$$

$$T(n) = 5T(n/5) + n$$

$$a = 5, \quad b = 5, \quad f(n) = n$$

$$n \log_b^a = n \log_5^5 = n^1 = n$$

$$f(n) = \Theta(n \log^a)$$

$$T(n) = \Theta(n \log n)$$

$$\textcircled{2} \quad T(n) = 2T(n/2) + n \log n$$

$$T(n) = 2T(n/2) + n \log n$$

$$a = 2, \quad b = 2, \quad f(n) = n \log n$$

$$n \log_b^a = n \log_2^2 = n$$

DATE: ___/___/___

DAY: ___/___/___

$$f(n) = \Theta(n \log n) = \Theta(n \log_b^2 \log n)$$

$$T(n) = \Theta(n \log^2 n).$$

$$n + (n)T^2 = (n)T \quad ①$$